

- 10 (a) Describe how reflected ultrasound pulses may be used to obtain diagnostic information about internal structures.

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..... [2]

- (b) (i) Define specific acoustic impedance of a medium.

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..... [2]

- (ii) Table 10.1 shows some data for water and for glass.

Table 10.1

	density / kg m^{-3}	speed of sound / m s^{-1}
water	1000	1420
glass	2500	4560

Determine the intensity reflection coefficient for ultrasound that is incident on a water–glass boundary.

intensity reflection coefficient = [3]

[Total: 7]





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